

Micro-Confidence

Daily Micro-Challenge App for Building Social Confidence

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This document contains the PRD, BRD, FSD, Technical Documentation, and User Documentation for the Micro-Confidence application.

Table of Contents

1. Submission Overview — Idea, Target Audience, User Flow, Hosted Link
2. Product Requirements Document (PRD)
3. Business Requirements Document (BRD)
4. Functional Specification Document (FSD)
5. Technical Documentation
6. User Documentation
7. Appendix — Database Schema, API Reference, Changelog

1. Submission Overview

1.1 Idea

Micro-Confidence is a mobile-first web application that helps users build social confidence through daily micro-challenges. Every day, users receive one small, actionable social courage challenge (e.g., 'Compliment a stranger,' 'Ask a question in a meeting,' 'Say no to something you don't want to do') that takes under 60 seconds to complete. Users mark challenges as done or skip them, earning points through a gamified Courage Score system. Over time, they build streaks, unlock achievements, and progress through confidence tiers from 'Shy Starter' to 'Confidence Legend.' The app transforms the abstract goal of 'becoming more confident' into concrete, trackable daily actions.

1.2 Target Audience

Micro-Confidence is designed for the following user segments:

- **Young professionals (22-35)** who want to improve networking and workplace communication skills
- **Introverts and shy individuals** looking for structured, low-pressure ways to practice social interaction
- **Students (18-25)** preparing for interviews, presentations, and professional environments
- **Remote workers** who feel their social skills have declined and want to rebuild confidence
- **Self-improvement enthusiasts** who enjoy gamified habit-tracking and personal growth apps
- **Therapists and coaches** who may recommend it as a supplementary tool for clients working on social anxiety

1.3 User Flow

The end-to-end user journey through the application:

Step	Screen	Action	Outcome
1	Onboarding	User sees 3-step intro carousel explaining the concept	User understands the app's purpose
2	Sign Up	User registers with email/password via Supabase Auth	Account created, profile auto-generated
3	Username	User picks a unique username	Profile personalized
4	Home / Today	User sees today's challenge card with emoji, text, category, difficulty	Clear, single daily action presented
5	Complete / Skip	User taps 'I Did It!' or 'Skip for today'	Points awarded (or streak reset), confetti animation plays

Step	Screen	Action	Outcome
6	Reflection	User optionally adds a short reflection note	Bonus points earned, journal entry saved
7	Stats	User views streak counter, courage score ring, tier badge, weekly heatmap	Progress is visible and motivating
8	Journal	User scrolls past challenge history with filters	Long-term progress and patterns are visible
9	Achievements	User sees locked/unlocked badges in a grid	Gamification drives continued engagement
10	Profile	User adjusts difficulty preference, toggles dark mode, shares progress	Personalized experience

1.4 Hosted Web App Link

Live App URL: <https://my-confidence-quest.lovable.app/>

2. Product Requirements Document (PRD)

2.1 Product Vision

Micro-Confidence transforms social confidence building from an overwhelming, abstract goal into a simple daily habit. By presenting one tiny, actionable challenge each day, the app removes decision fatigue and analysis paralysis. The gamified scoring system and streak mechanics leverage behavioral psychology to create a positive feedback loop that keeps users engaged and progressing.

2.2 Problem Statement

Millions of people struggle with social confidence but lack a structured, low-pressure way to practice. Existing solutions are either too intimidating (public speaking courses, therapy), too vague ('just put yourself out there'), or too passive (reading self-help books). There is no app that provides daily, micro-sized social challenges with progress tracking and gamification to make confidence-building feel achievable and rewarding.

2.3 Goals & Success Metrics

Goal	Metric	Target
User Retention	7-day retention rate	> 40%
Daily Engagement	Daily active users / Monthly active users	> 25%
Streak Engagement	Users with 7+ day streaks	> 20% of active users
Completion Rate	Challenges completed vs skipped	> 60% completion
Reflection Usage	Users who add reflections	> 30% of completions
Tier Progression	Users reaching 'Brave Beginner' tier	> 15% within 30 days

2.4 Feature Priority Matrix

Priority	Feature	Status
P0 (Must Have)	Daily challenge assignment + complete/skip	Built
P0 (Must Have)	Courage Score + Tier system	Built
P0 (Must Have)	Streak tracking with visual feedback	Built
P0 (Must Have)	User authentication (Supabase Auth)	Built
P1 (Should Have)	Reflection journaling	Built
P1 (Should Have)	Achievement badges system	Built

Priority	Feature	Status
P1 (Should Have)	Category-based challenge filtering	Built
P1 (Should Have)	Difficulty preference setting	Built
P2 (Nice to Have)	Social sharing cards	Built
P2 (Nice to Have)	Webhook integration (n8n)	Built
P2 (Nice to Have)	PWA installable on home screen	Built
P3 (Future)	Community leaderboard	Planned
P3 (Future)	Friend challenges	Planned
P3 (Future)	AI-personalized challenge selection	Planned

2.5 User Personas

Persona 1: Priya — The Quiet Professional

Age: 27 | Role: Software Developer | Location: Bangalore, India

Priya is great at her job but struggles to speak up in meetings. She wants to grow into a leadership role but feels held back by her reluctance to share ideas publicly. She needs a low-stakes way to practice assertiveness daily without feeling overwhelmed.

Persona 2: Alex — The Remote Worker

Age: 31 | Role: Marketing Manager | Location: Austin, USA

Alex has worked remotely for 3 years and notices their social skills have declined. Simple interactions like ordering coffee or making small talk now feel uncomfortable. They want a structured way to rebuild confidence through small daily actions.

Persona 3: Sam — The College Student

Age: 20 | Role: Computer Science Student | Location: London, UK

Sam has job interviews coming up and dreads networking events. They want to practice social skills in everyday situations first before tackling high-pressure scenarios. Gamification and streaks appeal to their competitive nature.

3. Business Requirements Document (BRD)

3.1 Business Objectives

- Build a scalable micro-challenge platform that helps users develop social confidence
- Achieve product-market fit with a free tier that drives organic growth through social sharing
- Establish a data pipeline (via n8n webhooks) for behavioral analytics and user insights
- Create a foundation for future monetization through premium challenge packs and coaching integrations
- Demonstrate a complete, production-ready full-stack application for portfolio/submission purposes

3.2 Stakeholders

Stakeholder	Role	Interest
Product Owner	Decision maker	Feature prioritization, roadmap alignment
End Users	Primary consumers	Engaging experience, measurable confidence growth
Developers	Engineering team	Clean architecture, maintainable codebase
Reviewers / Judges	Evaluators	Code quality, UX polish, feature completeness
Mental Health Advisors	Domain experts	Challenge appropriateness, user wellbeing

3.3 Business Rules

- Each user receives exactly one challenge per day, assigned at local midnight
- Challenges must not repeat within a 14-day window for the same user
- Difficulty scaling adapts to the user's courage score and preference setting
- All challenges must be completable in under 60 seconds of actual action
- User data is private by default (RLS enforced); no user can see another's data
- Webhook data transmission must be fire-and-forget — failures must never block the UI

3.4 Revenue Model (Future)

Tier	Price	Features
Free	\$0	1 daily challenge, basic stats, 5 categories, streaks
Pro	\$4.99/mo	Challenge rerolls, advanced analytics, custom categories, export data
Teams	\$9.99/mo per user	Team challenges, leaderboards, manager dashboard, API access

Tier	Price	Features
Enterprise	Custom	White-label, SSO, custom challenge libraries, coaching integration

3.5 Risk Assessment

Risk	Likelihood	Impact	Mitigation
User drops off after initial novelty	High	High	Streak mechanics, achievement system, difficulty progression
Challenge content feels repetitive	Medium	High	55+ seed challenges, 14-day no-repeat window, future AI generation
Social challenges cause anxiety	Medium	Medium	Difficulty settings, skip option, encouraging tone, no penalties
Data privacy concerns	Low	High	Supabase RLS, no cross-user data access, GDPR-compliant design
Webhook/n8n downtime	Low	Low	Fire-and-forget pattern, app works fully without webhook

4. Functional Specification Document (FSD)

4.1 Feature Specifications

4.1.1 Authentication System

Description: Email/password authentication powered by Supabase Auth with auto-profile creation.

Trigger: User clicks 'Get Started' on onboarding screen.

Flow:

- User enters email + password
- Supabase Auth creates auth.users record
- Database trigger *handle_new_user()* auto-creates a profiles row
- User is prompted to choose a username
- Username is validated for uniqueness and saved to profiles table

Validation Rules: Email must be valid format. Password minimum 6 characters. Username must be unique, 3-20 characters, alphanumeric + underscores only.

4.1.2 Daily Challenge Assignment

Description: Each day, the system assigns one challenge to each user based on their history and preferences.

Function: *get_or_assign_daily_challenge(p_user_id)*

Logic:

- Check if a daily_log exists for today — if yes, return it
- If no log exists, select a random challenge not seen in the last 14 days
- Weight selection by user's difficulty_preference and courage_score
- Insert new daily_log with status 'pending'
- Return challenge details as JSON

4.1.3 Scoring System

Function: *complete_challenge(p_user_id, p_log_id, p_reflection)*

Component	Formula	Example (Day 5, Difficulty 3, with Reflection)
Base Points	10 (fixed)	10
Streak Bonus	current_streak x 2	5 x 2 = 10
Difficulty Bonus	difficulty x 3	3 x 3 = 9
Reflection Bonus	5 (if reflection added)	5

Component	Formula	Example (Day 5, Difficulty 3, with Reflection)
Total	Sum of all components	34 points

4.1.4 Tier Progression

Score Range	Tier Name	Badge	Description
0 – 50	Shy Starter	Seedling	Just beginning the confidence journey
51 – 150	Quiet Rebel	Plant	Starting to push comfort zone boundaries
151 – 300	Brave Beginner	Fire	Consistently showing up with courage
301 – 500	Social Spark	Lightning	Confidence is becoming a visible habit
501 – 800	Courage Builder	Flexed Arm	Tackling harder challenges regularly
801 – 1200	Fearless Friend	Lion	Social confidence is second nature
1201+	Confidence Legend	Crown	Inspiring others with daily courage

4.2 Challenge Categories

Category	Count	Difficulty Range	Example Challenge
Conversation	12	1 – 4	Start a conversation with someone you've never spoken to
Kindness	11	1 – 3	Write a thank-you message to someone who helped you
Assertiveness	11	2 – 5	Say no to something without over-explaining
Vulnerability	10	2 – 4	Admit you don't know something and ask for help
Leadership	11	2 – 5	Give someone constructive feedback, kindly

4.3 Achievement Specifications

Achievement	Type	Requirement	Icon
First Step	Total	1 challenge completed	Seedling
Getting Started	Total	5 challenges completed	Walking
Double Digits	Total	10 challenges completed	Keycap 10
Century Club	Total	100 challenges completed	100

Achievement	Type	Requirement	Icon
On Fire	Streak	7-day streak	Fire
Two Week Warrior	Streak	14-day streak	Swords
Unstoppable	Streak	30-day streak	Rocket
Deep Thinker	Reflection	10 reflections added	Brain
Maximum Courage	Difficulty	Complete a difficulty 5 challenge	Lion
Rising Star	Score	Courage score reaches 100	Star
Brave Heart	Score	Courage score reaches 500	Purple Heart
Confidence Legend	Score	Courage score reaches 1200	Crown

5. Technical Documentation

5.1 System Architecture

Micro-Confidence follows a modern JAMstack architecture with a React frontend, Supabase Backend-as-a-Service (BaaS), and n8n for workflow automation. The frontend communicates with Supabase via the JavaScript client library, using RPC calls for complex operations and direct table queries for simple reads.

Layer	Technology	Purpose
Frontend	React 18 + TypeScript + Tailwind CSS + shadcn/ui	User interface, state management, routing
Animation	Framer Motion	Page transitions, micro-interactions, confetti
Auth	Supabase Auth	Email/password authentication, session management
Database	Supabase (PostgreSQL)	Data storage, RLS security, server-side functions
Automation	n8n (webhook-based)	Data logging, email reminders, weekly reports
Hosting	Lovable (Cloudflare)	Static site hosting, CDN, SSL, custom domain
Icons	Lucide React	Consistent iconography across the app

5.2 Database Schema

The application uses 5 PostgreSQL tables with Row Level Security (RLS) enabled on all:

Table	Primary Key	Key Columns	RLS Policy
profiles	id (UUID, FK auth.users)	username, courage_score, current_streak, tier	Users read/write own row only
challenges	id (UUID)	text, category, difficulty, emoji	All authenticated users can read
daily_logs	id (UUID)	user_id, challenge_id, date, status, reflection	Users read/write own logs only
achievements	id (UUID)	name, requirement_type, requirement_value	All authenticated users can read
user_achievements	id (UUID)	user_id, achievement_id, unlocked_at	Users read/write own badges only

5.3 Server-Side Functions (RPC)

Function	Parameters	Returns	Description
get_or_assign_daily_challenge	p_user_id (UUID)	JSON (challenge details)	Returns today's challenge or assigns a new one
complete_challenge	p_user_id, p_log_id, p_reflection	JSON (points breakdown)	Marks done, calculates score, updates profile
skip_challenge	p_user_id, p_log_id	JSON (status)	Marks skipped, resets streak
calculate_tier	score (INTEGER)	TEXT (tier name)	Pure function mapping score to tier

5.4 Webhook Integration (n8n)

After every challenge completion or skip, the app sends a POST request to the configured webhook URL. This is a fire-and-forget call wrapped in try/catch — webhook failures never block the UI.

Payload schema:

Field	Type	Description
event	string	'challenge_update'
user_id	UUID	Supabase user ID
username	string	User's chosen username
challenge_text	string	The challenge that was completed/skipped
category	string	conversation kindness assertiveness vulnerability leadership
difficulty	integer (1-5)	Challenge difficulty level
status	string	'completed' or 'skipped'
reflection	string null	User's optional reflection note
points_earned	integer	Points awarded (0 if skipped)
new_courage_score	integer	Updated total courage score
current_streak	integer	Current consecutive day streak
tier	string	Current tier name
timestamp	ISO 8601	Event timestamp

5.5 n8n Workflow Architecture

Workflow	Trigger	Pipeline	Output
1. Data Store	Webhook POST	Parse data → Enrich	Google Sheets row
2. Daily Reminder	Cron 6 PM IST	Query pending users → Filter	Email/Slack notification
3. Weekly Summary	Cron Mon 9 AM IST	Aggregate stats → Format	Summary email per user
4. Achievement Alert	Webhook POST	Parse → Classify importance	Slack post + email + log

5.6 Security Measures

- **Row Level Security (RLS):** Enabled on all 5 tables; users can only access their own data
- **Auth:** Supabase Auth with JWT tokens; anon key is public-safe (RLS enforces access)
- **HTTPS:** Enforced via Lovable/Cloudflare hosting with auto-provisioned SSL
- **Input Validation:** Username uniqueness, email format, password strength checked server-side
- **Webhook Security:** Fire-and-forget; no sensitive data exposed beyond user_id and username
- **CORS:** Configured via Supabase project settings to allow only the app domain

6. User Documentation

6.1 Getting Started

Step 1: Visit the app at your hosted URL or install it on your home screen (PWA supported).

Step 2: Tap 'Get Started' and create an account with your email and a password.

Step 3: Choose a username — this is how you'll be identified in the app.

Step 4: You'll see your first daily challenge. Read it, take action, and tap 'I Did It!'

6.2 Daily Challenge

Every day you'll receive one micro-challenge designed to stretch your social comfort zone just a little. Challenges are categorized into five areas: Conversation, Kindness, Assertiveness, Vulnerability, and Leadership. Each has a difficulty rating from 1 (easy) to 5 (hard).

To complete: Tap the green 'I Did It!' button after you've done the challenge in real life.

To skip: Tap 'Skip for today' if you're not feeling it — no judgment, but your streak resets.

Add a reflection: Optionally write a short note about how it went for +5 bonus points.

6.3 Understanding Your Score

Your Courage Score is calculated from four components:

- **Base points:** 10 points for every completed challenge
- **Streak bonus:** 2 extra points per consecutive day (Day 10 = +20 bonus)
- **Difficulty bonus:** Harder challenges earn more (difficulty x 3 points)
- **Reflection bonus:** +5 points when you add a reflection

Your score determines your tier — from Shy Starter (0-50) all the way to Confidence Legend (1201+).

6.4 Streaks

A streak counts consecutive days where you completed (not skipped) your challenge. Streaks are powerful motivators and unlock achievements at 7, 14, and 30 days. Skipping a day resets your streak to 0, but your total score and tier are never lost.

6.5 Achievements

Achievements are badges you unlock by hitting milestones. They appear as a grid on the Achievements screen — locked badges are greyed out, unlocked ones glow with color. There are 15 achievements across categories: total challenges completed, streak milestones, reflection count, difficulty feats, and score thresholds.

6.6 Settings & Customization

- **Difficulty Preference:** Choose 'easier', 'balanced', or 'harder' to influence which challenges you receive
- **Dark Mode:** Toggle between light and dark themes
- **Share Progress:** Share your completed challenge as a text card on social media
- **Sign Out:** Securely log out from your account

6.7 FAQ

Q: Can I change today's challenge?

A: Currently no — the constraint of 'one challenge, no rerolls' is intentional to build commitment.

Q: What happens if I miss a day?

A: Your streak resets to 0, but your total score and tier remain. You can always start a new streak tomorrow.

Q: Is my data private?

A: Yes. Row Level Security ensures no user can see another user's data. Your challenges, reflections, and scores are visible only to you.

Q: Can I use this on my phone?

A: Yes! The app is mobile-first and can be installed as a PWA on your home screen for an app-like experience.

7. Appendix

7.1 Environment Variables

Variable	Description	Example
VITE_SUPABASE_URL	Supabase project URL	https://abc123.supabase.co
VITE_SUPABASE_ANON_KEY	Supabase anonymous/public key	eyJhbGciOi...
VITE_WEBHOOK_URL	n8n webhook for challenge events	https://neelswap.app.n8n.cloud/webhook/micro-confidence-webhook
VITE_ACHIEVEMENT_WEBHOOK_URL	n8n webhook for achievement events	https://neelswap.app.n8n.cloud/webhook/micro-confidence-achievement

7.2 Technology Versions

Technology	Version	Purpose
React	18.x	UI framework
TypeScript	5.x	Type-safe JavaScript
Tailwind CSS	3.x	Utility-first CSS framework
shadcn/ui	Latest	Accessible UI component library
Framer Motion	11.x	Animation library
Supabase JS	2.x	Backend client library
Lucide React	0.383+	Icon library
n8n	Cloud / 1.x	Workflow automation

7.3 Changelog

Version	Date	Changes
1.0.0	2026-08-25	Initial release: all 6 screens, 55 challenges, 15 achievements, 4 n8n workflows, Supabase integration, PWA support